EST I — LITERACY & MATH • UNIT 0.01

EST I Literacy & Quantitative Diagnostic Drill

ANSWER KEY & SOLUTIONS

No-Calculator & Calculator Readiness • Official EST I Curriculum

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 D	Q3 B	Q4 B	Q5 B
Q6 B	Q7 C	Q8 A	Q9 D	Q10 A
Q11 A	Q12 A	Q13 C	Q14 A	Q15 A
Q16 C	Q17 D	Q18 A	Q19 C	Q20 C
Q21 B	Q22 A	Q23 C	Q24 D	Q25 D

EST I Literacy & Quantitative Diagnostic Drill Practice Exercises

1 MCQ

Consider the functions $f(x) = 2x + 3$ and $g(x) = x + 7$. At what x-value do the graphs of $f(x)$ and $g(x)$ intersect?

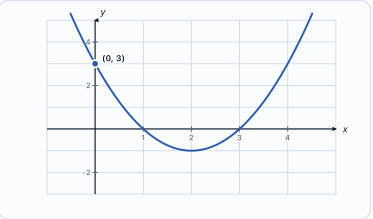
- **A** 4 ✓
- **B** 7
- **C** 3
- **D** 10

WORKED SOLUTION & VERIFICATION

To find the x-value where the graphs of $f(x)$ and $g(x)$ intersect, we set the two functions equal to each other: $f(x) = g(x)$. \n\nSubstituting the given expressions for $f(x)$ and $g(x)$: \n$2x + 3 = x + 7$ \n\nTo solve for x, subtract x from both sides of the equation: \n$2x - x + 3 = x - x + 7$ \n\nThis simplifies to: \n$x + 3 = 7$ \n\nNext, subtract 3 from both sides of the equation: \n$x + 3 - 3 = 7 - 3$ \n\nWhich gives the solution: \n$x = 4$ \n\nTherefore, the graphs of $f(x)$ and $g(x)$ intersect at $x = 4$.

2 MCQ

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. What is the value of $f(0)$?



- **A** -3
- **B** -1
- **C** 1
- **D** 3 ✓

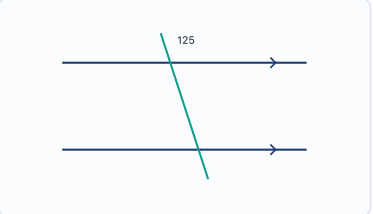
WORKED SOLUTION & VERIFICATION

To find the value of $f(0)$, we substitute $x = 0$ into the function's equation: $f(x) = x^2 - 4x + 3$. Therefore, $f(0) = (0)^2 - 4(0) + 3 = 0 - 0 + 3 = 3$. This corresponds to the y-intercept of the graph.

3 MCQ

In the diagram below, lines AB and CD are parallel. Line segment EF is a transversal intersecting AB at G and CD at H. If the measure of angle $\angle AGH$ is 125°, what is the measure of angle $\angle GHD$?

Diagram description: Two parallel horizontal lines AB and CD are intersected by a transversal line EF. Points A and C are on the left, B and D are on the right. G



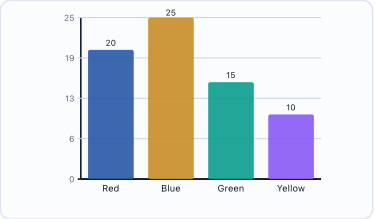
- **A** 55°
- **B** 125° ✓
- **C** 90°
- **D** 180°

WORKED SOLUTION & VERIFICATION

When two parallel lines are intersected by a transversal, consecutive interior angles are supplementary (add up to 180°). Angle $\angle AGH$ and angle $\angle GHC$ are consecutive interior angles. Therefore, $\angle AGH + \angle GHC = 180^\circ$.
We are given that $\angle AGH = 125^\circ$. So, $125^\circ + \angle GHC = 180^\circ$. Solving for $\angle GHC$, we get $\angle GHC = 180^\circ - 125^\circ = 55^\circ$.
Angle $\angle GHC$ and angle $\angle GHD$ are angles on a straight line (linear pair), so they are supplementary. Therefore, $\angle GHC + \angle GHD = 180^\circ$.
Substituting the value of $\angle GHC$, we get $55^\circ + \angle GHD = 180^\circ$.
Solving for $\angle GHD$, we get $\angle GHD = 180^\circ - 55^\circ = 125^\circ$.
Alternatively, angle $\angle AGH$ and angle $\angle GHD$ are alternate exterior angles. When two parallel lines are intersected by a transversal, alternate exterior angles are equal. Therefore, $\angle AGH = \angle GHD$. Since $\angle AGH = 125^\circ$, then $\angle GHD = 125^\circ$.

4 MCQ

The bar chart below shows the number of students who prefer each of four different colors. What is the total number of students who prefer either blue or green?



A 30

B **40** ✓

C 50

D 60

WORKED SOLUTION & VERIFICATION

The bar chart shows the following counts: Red: 20 students, Blue: 25 students, Green: 15 students, Yellow: 10 students. To find the total number of students who prefer either blue or green, we add the number of students who prefer blue to the number of students who prefer green.
Total = (Number of students who prefer blue) + (Number of students who prefer green)
Total = 25 + 15
Total = 40
Therefore, 40 students prefer either blue or green.

5 MCQ

A recipe calls for 2 cups of flour for every 3 cups of sugar. If 8 cups of flour are used, how many cups of sugar are needed?

A 10 cups

B **12 cups** ✓

C 11 cups

D 13 cups

WORKED SOLUTION & VERIFICATION

Let F be the amount of flour and S be the amount of sugar. The ratio of flour to sugar is given as $\frac{F}{S} = \frac{2}{3}$. We are given that $F = 8$ cups. We need to find S. We can set up the proportion: $\frac{8}{S} = \frac{2}{3}$. To solve for S, we can cross-multiply: $8 \times 3 = 2 \times S$. This simplifies to $24 = 2S$. Dividing both sides by 2, we get $S = \frac{24}{2} = 12$. Therefore, 12 cups of sugar are needed.

6 MCQ

If $9^{x+1} = 27^{x-1}$, what is the value of x?

A 2

B 5 ✓

C 3

D 7

WORKED SOLUTION & VERIFICATION

To solve the equation $9^{x+1} = 27^{x-1}$, express both sides in terms of a common base:
1. Recognize that both 9 and 27 are powers of 3:

$$9 = 3^2 \quad \text{and} \quad 27 = 3^3$$

2. Substitute these into the original equation:

$$(3^2)^{x+1} = (3^3)^{x-1}$$

3. Use the power of a power rule, $(a^m)^n = a^{mn}$, distributing across both terms in each exponent:

$$3^{2(x+1)} = 3^{3(x-1)}$$

$$3^{2x+2} = 3^{3x-3}$$

4. Since the bases are equal and positive (and not equal to 1), equate the exponents:

$$2x + 2 = 3x - 3$$

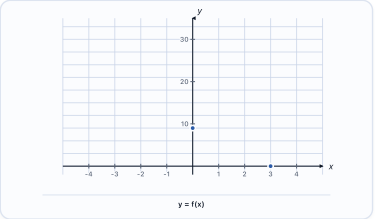
$$2 + 3 = 3x - 2x$$

$$x = 5$$

Common misconceptions:
- Forgetting to distribute the factor to the constant term (e.g., writing $2x + 1 = 3x - 1$) yields $x = 2$.
- Treating the bases as linear multipliers (e.g., $9(x + 1) = 27(x - 1)$) also incorrectly leads to $x = 2$.

7 MCQ

The graph of the quadratic function $y = x^2 - 6x + 9$ is a parabola. What is the x-coordinate of the vertex of this parabola?



A 0

B 6

C 3 ✓

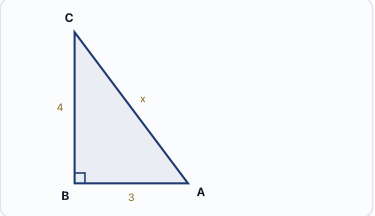
D -3

WORKED SOLUTION & VERIFICATION

The standard form of a quadratic function is $y = ax^2 + bx + c$. The x-coordinate of the vertex of a parabola given in this form is $x = -\frac{b}{2a}$. In the given function, $y = x^2 - 6x + 9$, we have $a = 1$ and $b = -6$. Therefore, the x-coordinate of the vertex is $x = -\frac{-6}{2(1)} = -\frac{-6}{2} = 3$.

8 MCQ

In the right-angled triangle ABC shown below, $\angle ABC = 90^\circ$, $AB = 3$, and $BC = 4$. What is the length of the hypotenuse AC?



A 5 ✓

B 7

C 12

D 25

WORKED SOLUTION & VERIFICATION

This question involves a direct application of the Pythagorean theorem in a right-angled triangle. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In triangle ABC, with $\angle ABC = 90^\circ$, AB and BC are the two shorter sides (legs), and AC is the hypotenuse.

According to the Pythagorean theorem:

$$AC^2 = AB^2 + BC^2$$

Given $AB = 3$ and $BC = 4$:

$$AC^2 = 3^2 + 4^2$$

$$AC^2 = 9 + 16$$

$$AC^2 = 25$$

To find the length of AC, we take the square root of both sides:

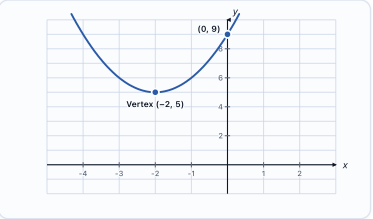
$$AC = \sqrt{25}$$

$$AC = 5$$

Therefore, the length of the hypotenuse AC is 5.

9 MCQ

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The vertex of the parabola is at the point $(-2, 5)$ and the parabola passes through the point $(0, 9)$. What is the value of b?



A -2

B 2

C -4

D 4 ✓

WORKED SOLUTION & VERIFICATION

The vertex form of a quadratic function is given by $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. In this case, the vertex is $(-2, 5)$, so $h = -2$ and $k = 5$. Thus, the function can be written as $f(x) = a(x - (-2))^2 + 5$, which simplifies to $f(x) = a(x + 2)^2 + 5$.

The parabola also passes through the point $(0, 9)$. We can substitute these values into the equation to solve for a:

$$9 = a(0 + 2)^2 + 5$$

$$9 = a(2)^2 + 5$$

$$9 = 4a + 5$$

$$4 = 4a$$

$$a = 1$$

Now we have the function in vertex form: $f(x) = (x + 2)^2 + 5$.

To find the value of b, we expand this expression to the standard form $f(x) = ax^2 + bx + c$:

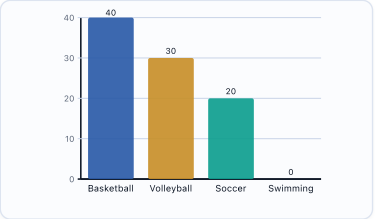
$$f(x) = (x^2 + 4x + 4) + 5$$

$$f(x) = x^2 + 4x + 9$$

Comparing this to $f(x) = ax^2 + bx + c$, we have $a = 1$, $b = 4$, and $c = 9$. The question asks for the value of b.

10 MCQ

The bar chart below shows the number of students who participated in different sports. The number of students who participated in swimming is unknown and represented by x. If the total number of students who participated in all sports is 120, and the number of students who participated in basketball, volleyball, and soccer are 40, 30, and 20 respectively, what is the value of x?



A 30 ✓

B 20

C 40

D 50

WORKED SOLUTION & VERIFICATION

The bar chart represents the number of students in different sports. Let the number of students in swimming be x.
The number of students who participated in basketball is 40.
The number of students who participated in volleyball is 30.
The number of students who participated in soccer is 20.
The total number of students who participated in all sports is 120.
To find the value of x, we can set up an equation where the sum of students in each sport equals the total number of students:

$$x + 40 + 30 + 20 = 120$$

Combine the known numbers:

$$x + 90 = 120$$

Subtract 90 from both sides to solve for x:

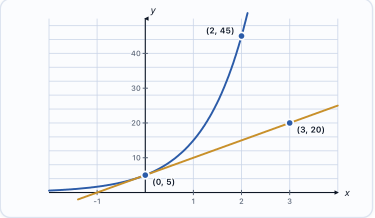
$$x = 120 - 90$$

$$x = 30$$

Therefore, the number of students who participated in swimming is 30.

11 MCQ

The diagram shows the graphs of the exponential function $y = a \cdot b^x$ and the linear function $y = mx + c$. The exponential function passes through the points $(0, 5)$ and $(2, 45)$. The linear function passes through the points $(0, 5)$ and $(3, 20)$. Find the value of a.



A 5 ✓

B 3

C 9

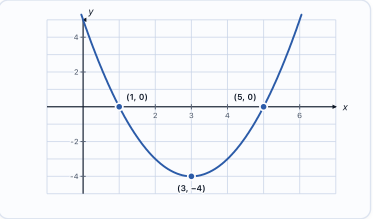
D 1

WORKED SOLUTION & VERIFICATION

For the exponential function $y = a \cdot b^x$:
1. Using the point $(0, 5)$:
$5 = a \cdot b^0$
Since $b^0 = 1$ for any non-zero b, we have $5 = a \cdot 1$, which means $a = 5$.
2. Using the point $(2, 45)$ and the value of $a = 5$:
$45 = 5 \cdot b^2$
Divide by 5:
$9 = b^2$
Taking the positive square root (as bases of exponential functions are typically positive), $b = 3$.
So, the exponential function is $y = 5 \cdot 3^x$.
For the linear function $y = mx + c$:
1. Using the point $(0, 5)$:
$5 = m(0) + c$
This gives $c = 5$.
2. Using the point $(3, 20)$ and $c = 5$:
$20 = m(3) + 5$
Subtract 5 from both sides:
$15 = 3m$
Divide by 3:
$m = 5$.
So, the linear function is $y = 5x + 5$.
The question asks for the value of a, which is the coefficient of the exponential term.

12 MCQ

The graph of the function $f(x) = ax^2 + bx + c$ is shown below. The parabola passes through the points $(1, 0)$ and $(5, 0)$. If the vertex of the parabola is at $(3, -4)$, what is the value of a?



- A 1 ✓
- B -1
- C 2
- D -2

WORKED SOLUTION & VERIFICATION

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. Since the parabola has x-intercepts at $(1, 0)$ and $(5, 0)$, we can write the function in factored form: $f(x) = a(x - 1)(x - 5)$. The x-coordinate of the vertex of a parabola in this form is the midpoint of the x-intercepts, which is $\frac{1+5}{2} = 3$. This matches the given x-coordinate of the vertex. We are also given that the vertex is at $(3, -4)$. We can substitute these coordinates into the factored form of the function to find the value of a:

$f(3) = a(3 - 1)(3 - 5)$

$-4 = a(2)(-2)$

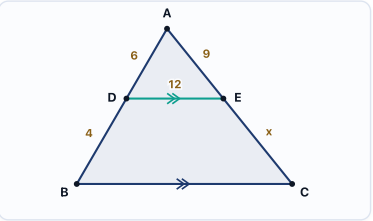
$-4 = -4a$

Dividing both sides by -4, we get:

$a = 1$.

13 MCQ

In the diagram below, line segment DE is parallel to line segment BC. If $AD = 6$, $DB = 4$, and $AE = 9$, what is the length of EC?



- A 7.2
- B 10
- C 6 ✓
- D 4.5

WORKED SOLUTION & VERIFICATION

Since DE is parallel to BC, by the Triangle Proportionality Theorem (also known as Thales's Theorem or the Basic Proportionality Theorem), the line DE divides the sides AB and AC proportionally.

This means that the ratio of AD to DB is equal to the ratio of AE to EC.

$$\frac{AD}{DB} = \frac{AE}{EC}$$

We are given the lengths: $AD = 6$, $DB = 4$, and $AE = 9$. We need to find EC.

Substitute the given values into the equation:

$$\frac{6}{4} = \frac{9}{EC}$$

To solve for EC, we can cross-multiply:

$$6 \times EC = 4 \times 9$$

$$6EC = 36$$

Now, divide by 6:

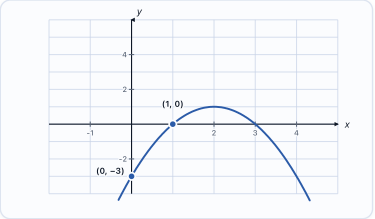
$$EC = \frac{36}{6}$$

$$EC = 6$$

Therefore, the length of EC is 6.

14 MCQ

The graph of the quadratic function $y = ax^2 + bx + c$ is shown below. The parabola intersects the x-axis at points $(1, 0)$ and $(p, 0)$, and has a y-intercept of $(0, -3)$. If the axis of symmetry of the parabola is the line $x = 2$, find the value of p.



- **A** 3 ✓
- **B** 4
- **C** 5
- **D** 6

WORKED SOLUTION & VERIFICATION

The roots of the quadratic equation $ax^2 + bx + c = 0$ are the x-intercepts of the parabola. We are given one root as $x_1 = 1$. The axis of symmetry of a parabola given by $y = ax^2 + bx + c$ is $x = -\frac{b}{2a}$. We are given that the axis of symmetry is $x = 2$.

For a quadratic function, the axis of symmetry is located exactly halfway between the roots. If the roots are x_1 and x_2, then the axis of symmetry is $x = \frac{x_1 + x_2}{2}$. We are given $x_1 = 1$ and the axis of symmetry $x = 2$. Let the other root be p. Therefore, we have:

$$2 = \frac{1 + p}{2}$$

Multiplying both sides by 2 gives:

$$4 = 1 + p$$

Subtracting 1 from both sides gives:

$$p = 3$$

We can also use the y-intercept to find the value of c. Since the parabola passes through $(0, -3)$, we have $y = a(0)^2 + b(0) + c = c$, so $c = -3$. The function can be written as $y = a(x - 1)(x - p)$. Substituting $(0, -3)$ and $p = 3$, we get:

$$-3 = a(0 - 1)(0 - 3)$$

$$-3 = a(-1)(-3)$$

$$-3 = 3a$$

$$a = -1$$

So the quadratic function is $y = -(x - 1)(x - 3) = -(x^2 - 4x + 3) = -x^2 + 4x - 3$. The axis of symmetry is $x = -\frac{4}{2(-1)} = -\frac{4}{-2} = 2$, which matches the given information. The roots are $x = 1$ and $x = 3$.

15 MCQ

If $i = \sqrt{-1}$, what is the value of i^{197}?

- **A** i ✓
- **B** 1
- **C** -1
- **D** $-i$

WORKED SOLUTION & VERIFICATION

The powers of the imaginary unit i repeat in a cycle of length 4:

$i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$.

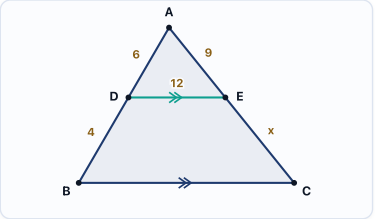
Divide the exponent 197 by 4:

$$197 = 4 \times 49 + 1$$

Thus, $i^{197} = (i^4)^{49} \times i^1 = (1)^{49} \times i^1 = i$.

16 MCQ

The diagram shows a triangle ABC, where D is a point on side AB and E is a point on side AC. It is given that DE is parallel to BC. If AD = 6 cm, DB = 4 cm, AE = 9 cm, and DE = 12 cm, find the length of the side BC.



A 24 cm

B 30 cm

C **20 cm** ✓

D 18 cm

WORKED SOLUTION & VERIFICATION

Since DE is parallel to BC, triangle ADE is similar to triangle ABC by the AA similarity criterion (angle A is common, and $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$ as corresponding angles).

For similar triangles, the ratio of corresponding sides is equal.

Therefore, we have:

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

We know AD = 6 cm and DB = 4 cm, so AB = AD + DB = 6 + 4 = 10 cm.

We are given AE = 9 cm and DE = 12 cm. Let BC = x cm.

Using the ratio of corresponding sides:

$$\frac{AD}{AB} = \frac{DE}{BC}$$

$$\frac{6}{10} = \frac{12}{x}$$

To solve for x, we can cross-multiply:

$$6x = 10 \times 12$$

$$6x = 120$$

$$x = \frac{120}{6}$$

$$x = 20$$

Therefore, the length of the side BC is 20 cm.

17 MCQ

A technician charges a flat initial fee of \$62 plus an hourly rate of \$21 for each hour of labor on a service call. The total charge C in dollars for h hours of work is given by $C = 21h + 62$. What is the best interpretation of the number 21 in this context?

A The number of hours needed to complete the service call

B The total cost for 62 hours of service

C The minimum flat fee charged for the visit, \$21

D **The hourly labor rate of \$21 per hour** ✓

WORKED SOLUTION & VERIFICATION

In the linear equation $C = 21h + 62$, the coefficient of h represents the slope or rate of change (cost per additional hour). The constant term 62 represents the C-intercept (initial flat fee when $h = 0$). Thus, 21 is the hourly labor rate in dollars per hour.

18 MCQ

A theater sells adult tickets for $\$14$ and child tickets for $\$10$. On a night when 12 tickets were sold for a total of $\$148$, how many adult tickets were sold?

A 7 ✓

B 6

C 8

D 5

WORKED SOLUTION & VERIFICATION

$$x + y = 12,\ 14x + 10y = 148 \Rightarrow x = 7,\ y = 5$$

19 MCQ

Data Set A has values clustered closely around its mean of 50, ranging from 48 to 52. Data Set B has the same mean of 50, but its values are widely dispersed, ranging from 20 to 80. Both sets have 100 observations. Which statement correctly compares their standard deviations, s_A and s_B?

A There is not enough information to compare the standard deviations without individual data values

B $s_A > s_B$, because Data Set A is more densely concentrated

C **$s_A < s_B$, because Data Set B has greater variability around the mean** ✓

D $s_A = s_B$, because both sets have identical means of 50

WORKED SOLUTION & VERIFICATION

Standard deviation measures the typical distance or spread of data values from their arithmetic mean.
Data Set A is tightly clustered within $[48, 52]$, so deviations $|x - 50|$ are small.
Data Set B has values spread out over $[20, 80]$, so deviations $|x - 50|$ are much larger.
Therefore, the standard deviation of Data Set A is strictly less than that of Data Set B: $s_A < s_B$.

20 MCQ

Consider the function $f(x)$ defined by $f(x) = |x - 2| + |x + 1|$. Find the minimum value of $f(x)$ and the interval over which this minimum is attained.

A Minimum value 2 on the interval $[-1, 2]$

B Minimum value 3 on the interval $(-\infty, -1]$

C **Minimum value 3 on the interval $[-1, 2]$** ✓

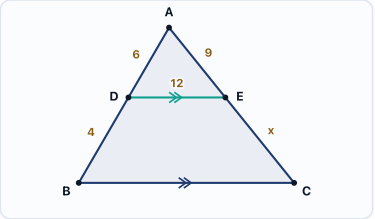
D Minimum value 2 on the interval $[2, \infty)$

WORKED SOLUTION & VERIFICATION

The function is defined as $f(x) = |x - 2| + |x + 1|$. We can analyze this function by considering the critical points where the expressions inside the absolute values change sign, which are $x = 2$ and $x = -1$. This divides the number line into three intervals:
Case 1: $x < -1$
In this interval, $(x - 2)$ is negative and $(x + 1)$ is negative. So, $f(x) = -(x - 2) - (x + 1) = -x + 2 - x - 1 = -2x + 1$. This is a decreasing linear function.
Case 2: $-1 \leq x \leq 2$
In this interval, $(x - 2)$ is non-positive and $(x + 1)$ is non-negative. So, $f(x) = -(x - 2) + (x + 1) = -x + 2 + x + 1 = 3$. This is a constant function.
Case 3: $x > 2$
In this interval, $(x - 2)$ is positive and $(x + 1)$ is positive. So, $f(x) = (x - 2) + (x + 1) = x - 2 + x + 1 = 2x - 1$. This is an increasing linear function.
Combining these cases, we see that:
- For $x < -1$, $f(x)$ decreases.
- For $-1 \leq x \leq 2$, $f(x) = 3$.
- For $x > 2$, $f(x)$ increases.
Therefore, the minimum value of $f(x)$ is 3, and this minimum value is attained for all x in the interval $[-1, 2]$.
Common misconception: Students might incorrectly assume the minimum occurs at one of the critical points and not consider the interval where the function is constant. For example, if they only check $x = -1$, they might get $f(-1) = |-1 - 2| + |-1 + 1| = |-3| + |0| = 3$. If they only check $x = 2$, they might get $f(2) = |2 - 2| + |2 + 1| = |0| + |3| = 3$. However, they must also confirm that the function does not dip below 3 in any other interval. The analysis of the three cases shows this is indeed the minimum.

21 MCQ

In the given diagram, $\triangle ABC$ is inscribed in a circle. Points D and E lie on sides AB and AC respectively. It is given that $AD = 6$, $DB = 4$, $AE = 9$, and DE is parallel to BC. Furthermore, $DE = 12$. Calculate the length of BC.



A 24

B 20 ✓

C 27

D 18

WORKED SOLUTION & VERIFICATION

Since $DE \parallel BC$, by the property of similar triangles (Thales's Theorem or Basic Proportionality Theorem), $\triangle ADE \sim \triangle ABC$.
The ratio of corresponding sides in similar triangles is equal. Therefore, we have:

$$\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$$

We are given $AD = 6$ and $DB = 4$. So, $AB = AD + DB = 6 + 4 = 10$.
We are given $AE = 9$ and $DE = 12$.
Using the ratio of similarity:

$$\frac{AD}{AB} = \frac{DE}{BC}$$

Substitute the known values:

$$\frac{6}{10} = \frac{12}{BC}$$

Now, solve for BC:

$$6 \times BC = 10 \times 12$$

$$6 \times BC = 120$$

$$BC = \frac{120}{6}$$

$$BC = 20$$

22 MCQ

Which of the following is equivalent to the expression $\frac{x^2-10x+24}{x^2-36}$ for all $x \neq \pm 6$?

A $\frac{x-4}{x+6}$ ✓

B $\frac{-10x+24}{-36}$

C $\frac{x+4}{x-6}$

D $\frac{x-4}{x-6}$

WORKED SOLUTION & VERIFICATION

Factor both the numerator and the denominator completely:
Numerator: $x^2 - 10x + 24 = (x - 6)(x - 4)$.
Denominator: $x^2 - 36 = (x - 6)(x + 6)$ (difference of two squares).
Dividing out the common factor $(x - 6)$:

$$\frac{(x-6)(x-4)}{(x-6)(x+6)} = \frac{x-4}{x+6}$$

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23 MCQ

A researcher is studying the relationship between the number of hours spent studying for a statistics exam and the score achieved. A random sample of 10 students yielded the following data, where x is the number of hours studied and y is the exam score:

The researcher models the relationship using a linear regression line of the form $y = \hat{a}x + \hat{b}$. Calculate the predicted exam score for a student who studies for $x = 7.5$ hours, given that the calculated regression line is $y = 5x + 60$.

Hours Studied (x)	Exam Score (y)
2	65
3	70
4	78
5	82
6	88
7	91
8	95
9	98
10	100
11	103

A. 92.5

B. 95.0

C. 97.5 ✓

D. 100.0

WORKED SOLUTION & VERIFICATION

The problem provides a linear regression model $\hat{y} = 5x + 60$, where x is the number of hours studied and $\hat{y}$ is the predicted exam score. We are asked to find the predicted exam score for a student who studies for $x = 7.5$ hours.

To find the predicted exam score, we substitute $x = 7.5$ into the regression equation:

$$\hat{y} = 5(7.5) + 60$$

First, calculate the product of 5 and 7.5:

$$5 \times 7.5 = 37.5$$

Now, add 60 to this result:

$$\hat{y} = 37.5 + 60$$

$$\hat{y} = 97.5$$

Therefore, the predicted exam score for a student who studies for 7.5 hours is 97.5.

Common errors could include incorrect multiplication of decimals (e.g., $5 \times 7.5 = 35$ or 37.0) leading to incorrect predictions, or errors in adding the intercept (e.g., $37.5 + 60 = 90.5$). A student might also confuse the predictor x with the predicted value $\hat{y}$ or try to interpolate directly from the data points without using the regression line. For instance, noticing that 7.5 hours is exactly halfway between 6 and 8 hours, a student might incorrectly average the scores for 6 and 8 hours ($(90 + 100)/2 = 95$), which is a linear interpolation but not the prediction from the calculated regression line.

24 MCQ

Which of the following is equivalent to $\log_b\left(\frac{x^5y^3}{z^4}\right)$ for all positive values of x, y, z?

A. $5\log_b(x) \cdot 3\log_b(y) - 4\log_b(z)$

B. $\frac{5\log_b(x)+3\log_b(y)}{4\log_b(z)}$

C. $5\log_b(x) - 3\log_b(y) - 4\log_b(z)$

D. $5\log_b(x) + 3\log_b(y) - 4\log_b(z)$ ✓

WORKED SOLUTION & VERIFICATION

Apply the fundamental logarithmic laws:

1) Quotient Rule: $\log_b(M/N) = \log_b(M) - \log_b(N)$

$$\log_b\left(\frac{x^5y^3}{z^4}\right) = \log_b(x^5y^3) - \log_b(z^4)$$

2) Product Rule: $\log_b(MN) = \log_b(M) + \log_b(N)$

$$= \log_b(x^5) + \log_b(y^3) - \log_b(z^4)$$

3) Power Rule: $\log_b(M^k) = k\log_b(M)$

$$= 5\log_b(x) + 3\log_b(y) - 4\log_b(z)$$

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25 MCQ

How many different integers between 15 and 45 contain only digits from the following list: 1, 3, 5, 7, 9 ?

A. 16

B. 6

C. 5

D. 7 ✓

WORKED SOLUTION & VERIFICATION

Between 15 and 45 (exclusive): The tens digit must be 1 or 3 (as 2 and 4 are not in the list). For tens digit 1, units digit can be 7 or 9 (giving 17, 19 — numbers > 15), yielding 2 numbers. For tens digit 3, units digit can be 1, 3, 5, 7, or 9 (giving 31, 33, 35, 37, 39 — all < 45), yielding 5 numbers. Total count = $2 + 5 = 7$.